Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti |

AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Staff-level LLM engineer who **owned AgentProbe end-to-end** - production LangChain/LangGraph architecture, RAG pipeline, 5-metric evaluation engine, multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), FastAPI backend, and React dashboard. Built it as a public open-source tool that other engineers adopt. 8+ years production Python; full-stack delivery at fintech scale; applied PyTorch + OpenAI API to automate 1.6M-record financial data workflows at Entrata. Targeting applied LLM engineering in fintech - where credit, risk, and intelligent decisioning intersect with the GenAI stack. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).
github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

LLM Application Architecture | LangChain / LangGraph | RAG + Vector Search | Production Eval Engine | Prompt Engineering | Multi-Provider LLM Routing | FastAPI / Python | React (Full-Stack) | LLM-as-Judge Design | AWS / Docker / CI/CD | Applied ML in FinTech | Cross-Functional Delivery

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Applied **Python, PyTorch, and OpenAI API** to automate HUD income limit data transformation, replacing manual Excel workflows across **1.6M records** - production applied ML in a fintech/compliance context.
- Built and delivered **20+ full-stack modules** (Python, TypeScript, PostgreSQL, React-equivalent) across **120+ Agile sprints** for **100K+ monthly active users**; drove cross-functional alignment across Product, Engineering, and QA.
- Improved production query performance by **30%**; maintained **99.5% uptime** for 6+ months via TDD/BDD; built internal debugging tools adopted across multiple squads - reducing debugging time by **40%**.
- Migrated thousands of client documents to AWS S3 resolving **90% of legacy storage constraints**; encrypted 100% of resident PHI via Libsodium - zero security incidents.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Mentored 30+ students in Python and databases; translated complex ML/data concepts across technical and non-technical audiences - the cross-functional communication staff-level roles require.

PROJECTS

AgentProbe: Production LLM Application Platform OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Staff-level end-to-end ownership of a production LLM application - every architectural decision from greenfield to public open-source. **LangChain/LangGraph orchestration**: 3 agents (Orchestrator, RAG Tracer, Auto Fix) with plan-execute-react loop and custom tool calls. **RAG + vector search**: ChromaDB, embedding pipeline, retrieval quality scoring. **Multi-provider routing**: Anthropic, OpenAI, Groq, Ollama - dynamic selection based on performance. **Production eval engine**: 5-metric LLM-as-judge (factual accuracy, hallucination, retrieval, completeness, compliance) with regression tracking and CI/CD gates. **Full-stack**: 15-endpoint FastAPI backend + React observability dashboard. Public tool adopted by other engineers. *github.com/venkatalakshmiparimala/AgentProbe*

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · Permutation Importance · GitHub Pages

Full ML lifecycle: feature engineering → model selection (RF vs LR) → rigorous evaluation (MDI + Permutation Importance) → deployment → stakeholder dashboard. **85%+ accuracy.** Demonstrates the end-to-end applied ML discipline - not just training but systematic measurement and deployment.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

NLP pipeline mapping natural language queries to FHIR APIs via LLMs, BERT, and embeddings. HIPAA-compliant RBAC. **~60% retrieval time reduction.** *github.com/venkatalakshmiparimala/AI_on_FHIR*

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

LLM & AI: LangChain, LangGraph, Multi-Provider LLM (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Prompt Engineering, LLM-as-Judge, LLM Evaluation, FastAPI

ML & Modeling: PyTorch, TensorFlow, scikit-learn, Random Forest, Feature Engineering, Permutation Importance, NumPy, Pandas, BERT, spaCy, Hugging Face, Statistical Modeling

Full-Stack & Cloud: React, AWS (S3, EC2, RDS, Glue), Azure, PostgreSQL, MySQL, Docker, CI/CD (GitHub Actions), RabbitMQ

Process: TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum